

# 1) OSI Model

- The OSI model has 7 layers.
- Each layer performs a specific networking function.
- A real-world protocol or technology is shown next to each layer.

|                       |                  |
|-----------------------|------------------|
| <b>7 Application</b>  | HTTP             |
| <b>6 Presentation</b> | TLS              |
| <b>5 Session</b>      | RPC              |
| <b>4 Transport</b>    | TCP              |
| <b>3 Network</b>      | IP               |
| <b>2 Data Link</b>    | Ethernet         |
| <b>1 Physical</b>     | Wi-Fi / Ethernet |

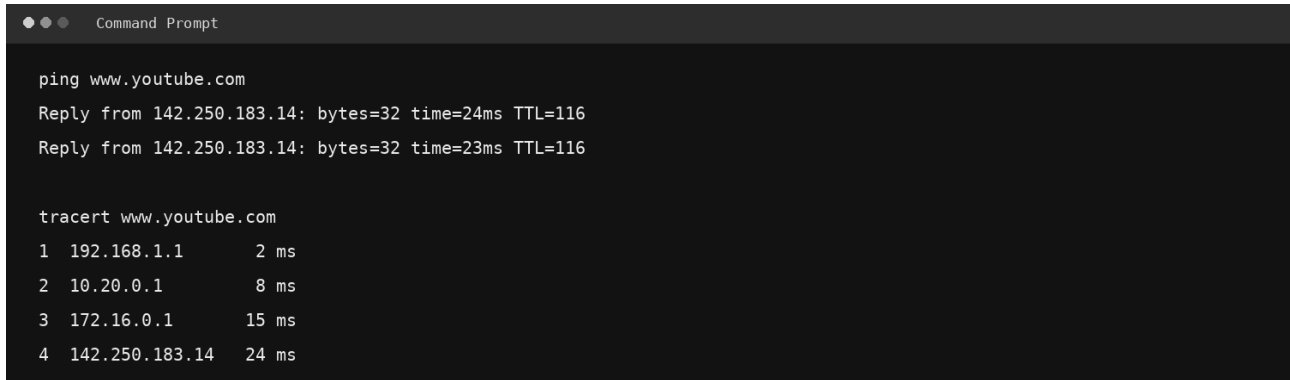
## 2) Apps and Network Protocols

- WhatsApp — HTTPS and TCP.
- Instagram — HTTPS and DNS.
- Zomato — HTTPS and DNS.
- Flipkart — HTTPS and TCP.
- Paytm — HTTPS and TLS.

| No. | App       | Protocol 1 | Protocol 2 |
|-----|-----------|------------|------------|
| 1   | WhatsApp  | HTTPS      | TCP        |
| 2   | Instagram | HTTPS      | DNS        |
| 3   | Zomato    | HTTPS      | DNS        |
| 4   | Flipkart  | HTTPS      | TCP        |
| 5   | Paytm     | HTTPS      | TLS        |

### 3) Ping and Tracert

- ping checks reachability and measures response time.
- tracert shows the path through intermediate network hops.
- OSI layers involved include Application, Transport, Network, Data Link and Physical.
- ICMP is used for network diagnostic messages.

A screenshot of a Windows Command Prompt window with a dark background. The title bar at the top shows three colored circles (red, yellow, green) and the text "Command Prompt". The command prompt shows the execution of two network diagnostic commands. The first command is "ping www.youtube.com", followed by two lines of output: "Reply from 142.250.183.14: bytes=32 time=24ms TTL=116" and "Reply from 142.250.183.14: bytes=32 time=23ms TTL=116". The second command is "tracert www.youtube.com", followed by four lines of output showing the path: "1 192.168.1.1 2 ms", "2 10.20.0.1 8 ms", "3 172.16.0.1 15 ms", and "4 142.250.183.14 24 ms".

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Command Prompt

ping www.youtube.com
Reply from 142.250.183.14: bytes=32 time=24ms TTL=116
Reply from 142.250.183.14: bytes=32 time=23ms TTL=116

tracert www.youtube.com
1 192.168.1.1      2 ms
2 10.20.0.1       8 ms
3 172.16.0.1      15 ms
4 142.250.183.14  24 ms
```

## 4) WhatsApp Message Through the OSI Model

- Application: the message is created in WhatsApp.
- Presentation: data is formatted and protected.
- Session: communication is maintained.
- Transport: TCP handles reliable transport.
- Network: IP addresses packets toward the destination.
- Data Link: packets are placed into frames.
- Physical: bits travel through Wi-Fi or mobile network.
- The receiving device moves back up the layers until the message appears.

|                     |                             |
|---------------------|-----------------------------|
| <b>Application</b>  | Message created in WhatsApp |
| <b>Presentation</b> | Data formatted / protected  |
| <b>Session</b>      | Communication maintained    |
| <b>Transport</b>    | TCP handles transport       |
| <b>Network</b>      | IP addresses packets        |
| <b>Data Link</b>    | Frames prepared             |
| <b>Physical</b>     | Bits sent over Wi-Fi/mobile |